

Frank Hou

frankhou@utexas.edu | linkedin.com/in/yufanhou | github.com/frankhou42

EDUCATION

University of Texas at Austin | *B.S. Computer Science, GPA: 3.7/4.0*

Expected May 2027

– Coursework: Operating Systems, Algorithms, Machine Learning, Data Structures, Computer Architecture, Linear Algebra

EXPERIENCE

Amazon | *Software Engineering Intern, Shipping*

May 2026 – Present

- Shifted Delivery Windows lookups from live DynamoDB reads to versioned S3 snapshots, improving low-latency schedule serving while reducing dependence on hot production paths
- Built AWS Glue and Spark ETL jobs scanning 2M+ delivery records every 30 minutes, validating snapshot freshness, and publishing durable read-optimized datasets to S3 for downstream services
- Implemented Java/Spring loaders with metadata validation, atomic in-memory swaps, stale-read guards, and rollback-safe snapshot promotion across service hosts without blocking traffic
- Deployed CDK, IAM, AppConfig, CodePipeline, CloudWatch alarms, and cross-account S3 permissions to support secure multi-stage production releases with rollback visibility

Scale AI | *GenAI / AI Safety Evaluation Intern*

May 2025 – May 2026

- Created prompts, reference solutions, and grading rubrics for 120+ Olympiad-level coding tasks, improving reliability and consistency of AI model evaluation across reviewers
- Built deterministic Python suites with 1,200+ unit, edge, and regression tests, reducing ambiguous grading decisions by 35% across repeated benchmark workflows
- Supported 10k+ weekly evaluations by contributing to React, TypeScript, Flask, AWS, and Kubernetes tooling while writing failure analyses that guided QA and model iteration

Machine Intelligence Through Decision-Making Lab | *Research Assistant*

Fall 2025 – Present

- Extended VIP and DERAILE offline RL frameworks in PyTorch to analyze 100k+ Franka-Kitchen and D4RL trajectories for model-based planning and ranking research
- Benchmarked value functions across 6 settings, exposing score collapse, high-variance rankings, weak action sensitivity, and trajectory-selection failures in candidate rollouts
- Built diagnostics for 50+ planning runs to visualize value landscapes, rollout quality, reward alignment, and learned dynamics failure modes during trajectory search

PROJECTS

Tree | *Local LLM Infrastructure System* | Swift, Ollama, LangChain

Fall 2025 – Spring 2026

- Built a macOS LLM context system with 300+ saved branches, reusable context paths, structured prompt workflows, and local-first AI experimentation for repeated tasks
- Integrated Llama 3.2 via Ollama for private on-device inference, reducing repeated cloud inference cost to \$0 while enabling fast iteration without external services
- Ran 100+ context-depth experiments and improved response stability by 18% through reusable context patterns and controlled prompt workflows across branches

MoodScribe | *AI Messaging Assistant* | PyTorch, React, Flask, TypeScript

Fall 2025

- Built a full-stack AI messaging assistant used by 100+ users and 2,000+ generated messages through React onboarding flows, Flask inference APIs, and PyTorch response generation
- Improved human-rated tone consistency by 22% by evaluating 300+ conversation samples, identifying personalization failures, and refining prompt strategies
- Reduced fine-tuning memory usage by 70% using LoRA adapters, FlashAttention, and Unsloth, enabling lightweight model adaptation on limited compute resources

TECHNICAL SKILLS

Languages: Python, Java, C/C++, TypeScript, JavaScript, Swift, SQL

AI/ML: PyTorch, TensorFlow, JAX, scikit-learn, NumPy, Pandas, LangChain, Offline RL, LLM Evaluation

Infrastructure/Systems: AWS, GCP, Kubernetes, Docker, Linux, CDK, DynamoDB, S3, Spark, CI/CD, Distributed Systems

Backend/Tools: Spring, Flask, Node.js, REST APIs, Git, CloudWatch, AppConfig, CodePipeline, IAM, Testing, Debugging

LEADERSHIP

Texas ACM | *Chair of Internal Affairs*

Fall 2023 – Present

- Led operations for 10+ technical and recruiting events by coordinating logistics, speaker planning, and member engagement

Sanger Learning Center | *Computer Science Tutor*

Fall 2025 – Present

- Tutored students weekly in data structures, algorithms, recursion, object-oriented programming, debugging, and core CS problem solving
- Explained Java, Python, runtime tradeoffs, and test-driven debugging strategies through guided sessions that built independent problem-solving habits